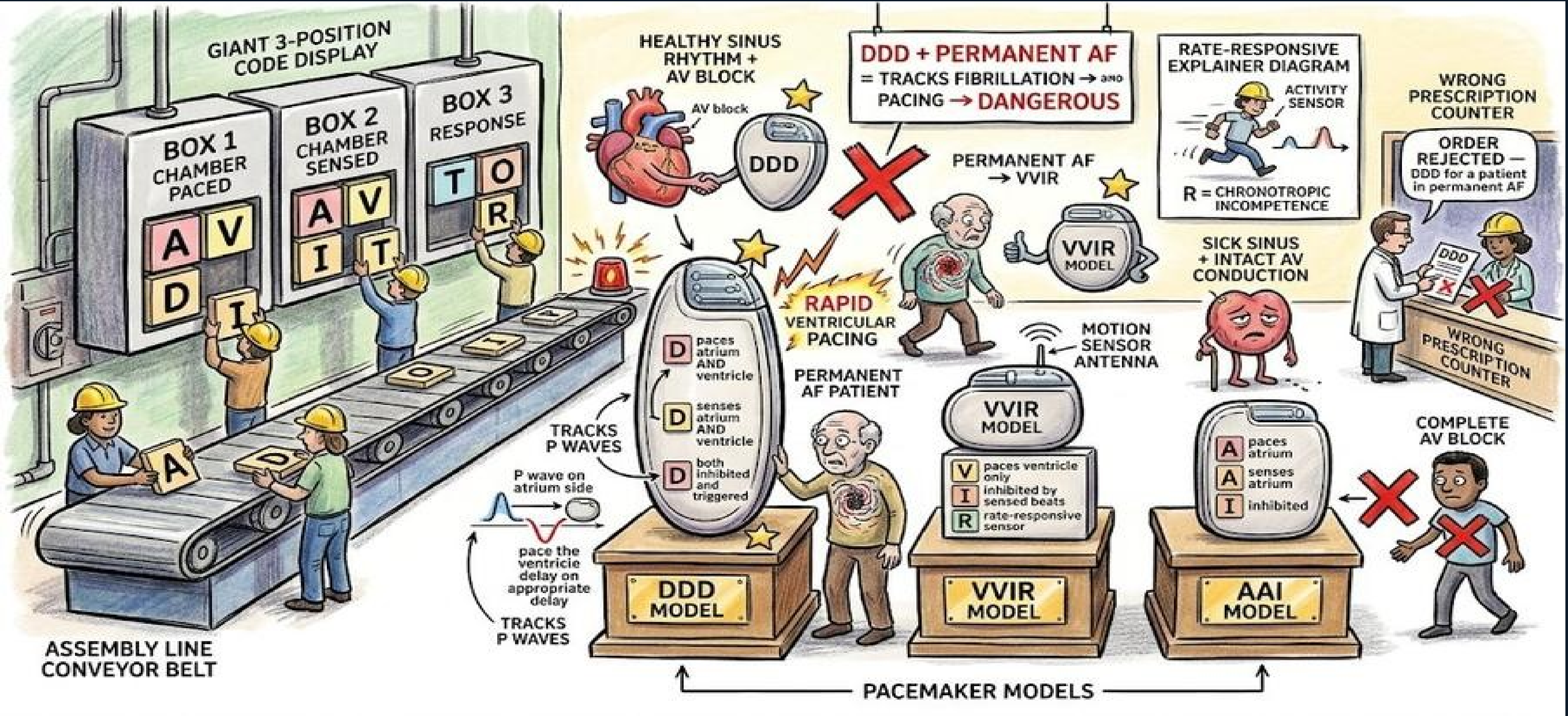
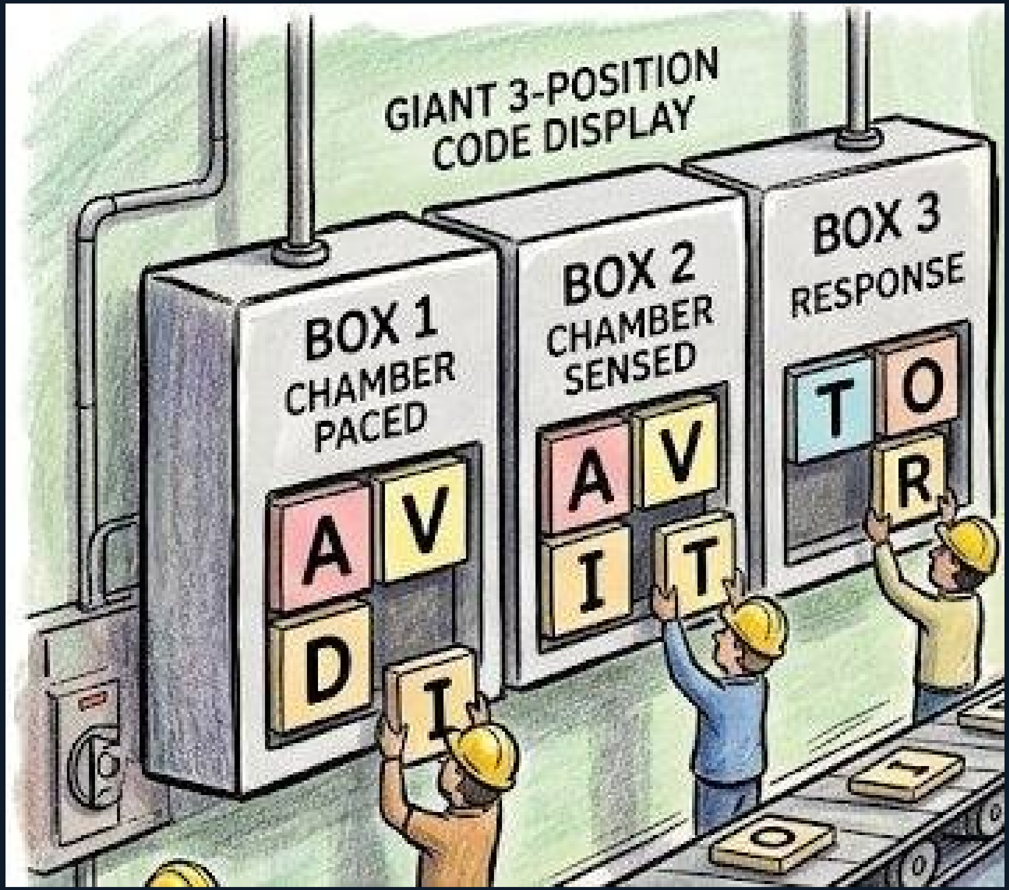
Scene H: Pacemaker Codes



Coding Factory · IMPORTANT · 2 Board Questions

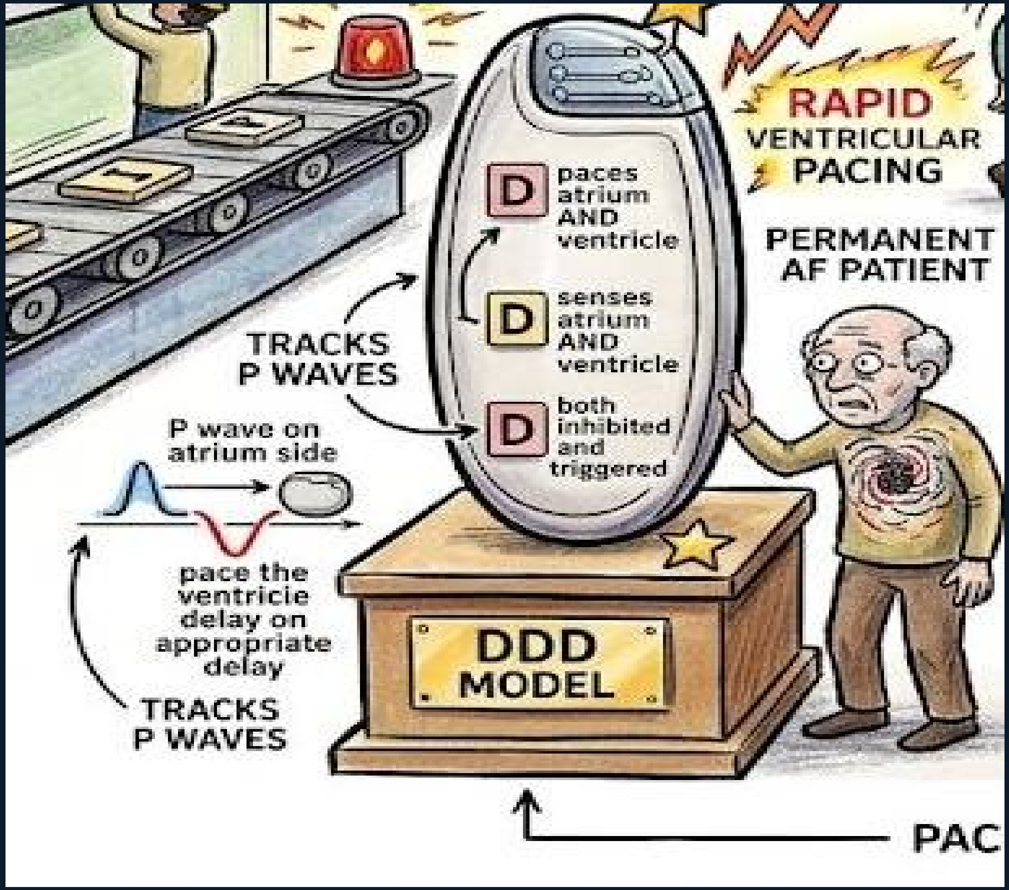


The 3-Position Code

A giant assembly line display shows three boxes: BOX 1 (CHAMBER PACED), BOX 2 (CHAMBER SENSED), BOX 3 (RESPONSE). Workers slot letter tiles: A, V, D, I, T, O, R.

Pacemaker code positions: Position 1 = what is paced (A=atrium, V=ventricle, D=dual). Position 2 = what is sensed (same letters). Position 3 = how the pacemaker responds to a sensed event (I=inhibited, T=triggered, D=both, O=none). Optional 4th letter R = rate-responsive.

Board example: DDD = paces atrium and ventricle, senses both, responds with inhibition or triggering as appropriate — the most physiologic mode.

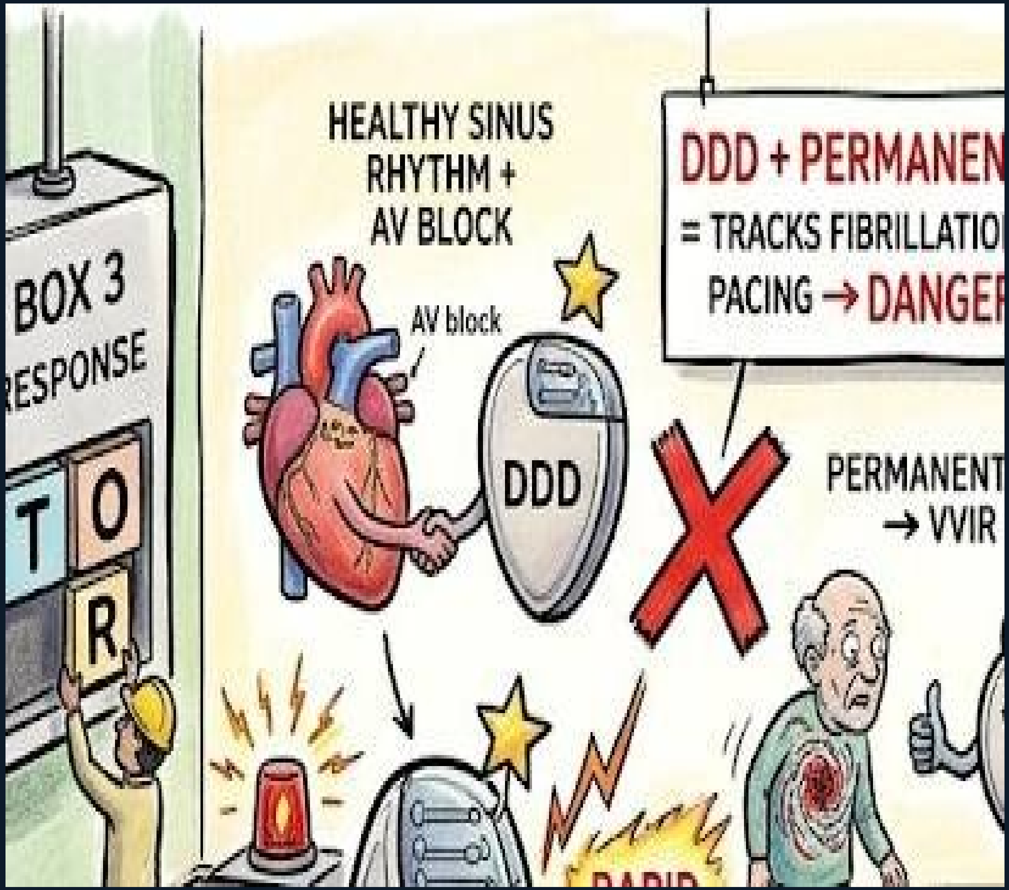


DDD Model — Most Physiologic

The DDD pacemaker on its pedestal shows: D = paces atrium AND ventricle, D = senses both, D = responds with inhibition AND triggering. It 'tracks P waves' — a sensed P wave triggers ventricular pacing after the appropriate AV delay.

DDD is ideal for patients with AV block in sinus rhythm — it maintains AV synchrony. The pacemaker waits for the atrial P wave, then paces the ventricle after an appropriate PR interval if no native conduction occurs.

Board trap: 'VVIR for a patient in sinus rhythm with complete AV block who is young and active' — suboptimal. DDD preserves AV synchrony and physiologic rate response.

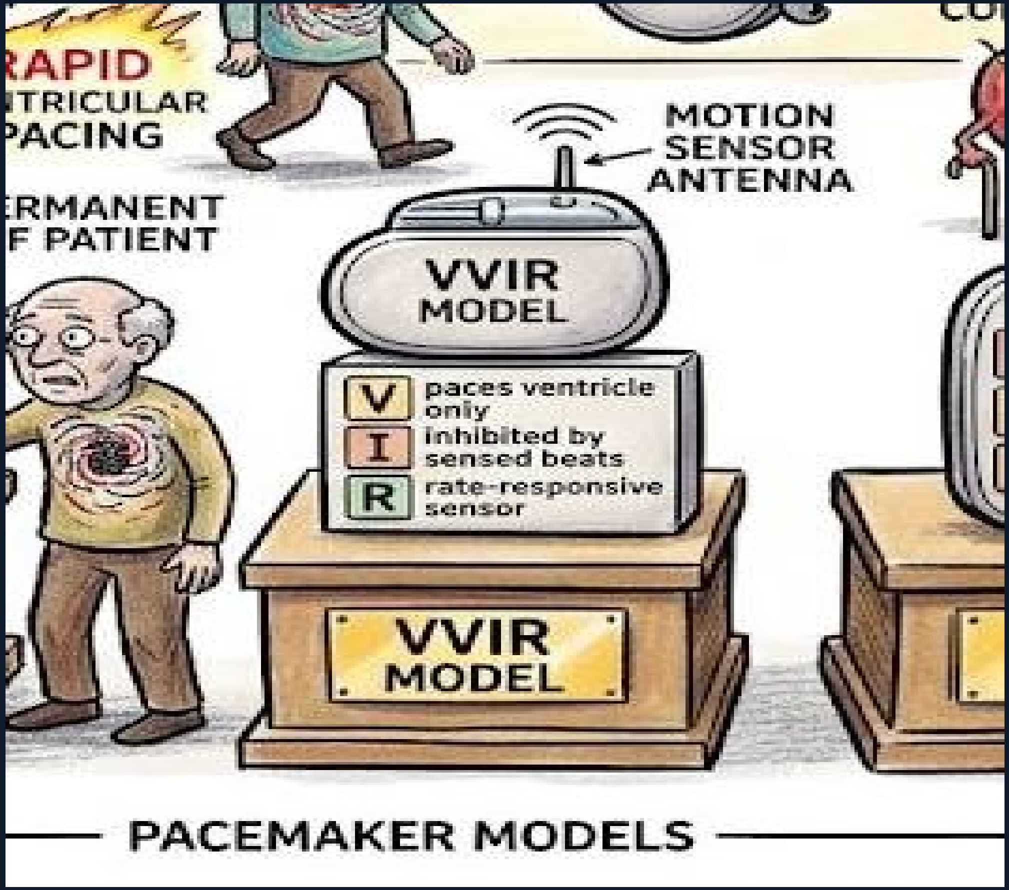


DDD + Permanent AF = Dangerous

A permanent AF patient approaches the DDD pacemaker. Immediately: the conveyor belt goes haywire, alarm blares, 'RAPID VENTRICULAR PACING' sparks erupt. Red X on the DDD device. Sign: 'DDD + PERMANENT AF = TRACKS FIBRILLATION → DANGEROUS.'

In permanent AF there are no organized P waves — only fibrillatory atrial activity at 350–600 impulses per minute. If a DDD pacemaker is programmed, it will attempt to track these atrial signals and pace the ventricle at dangerously high rates.

Board trap: choosing DDD for a patient with permanent AF and complete heart block. The correct mode is VVIR.

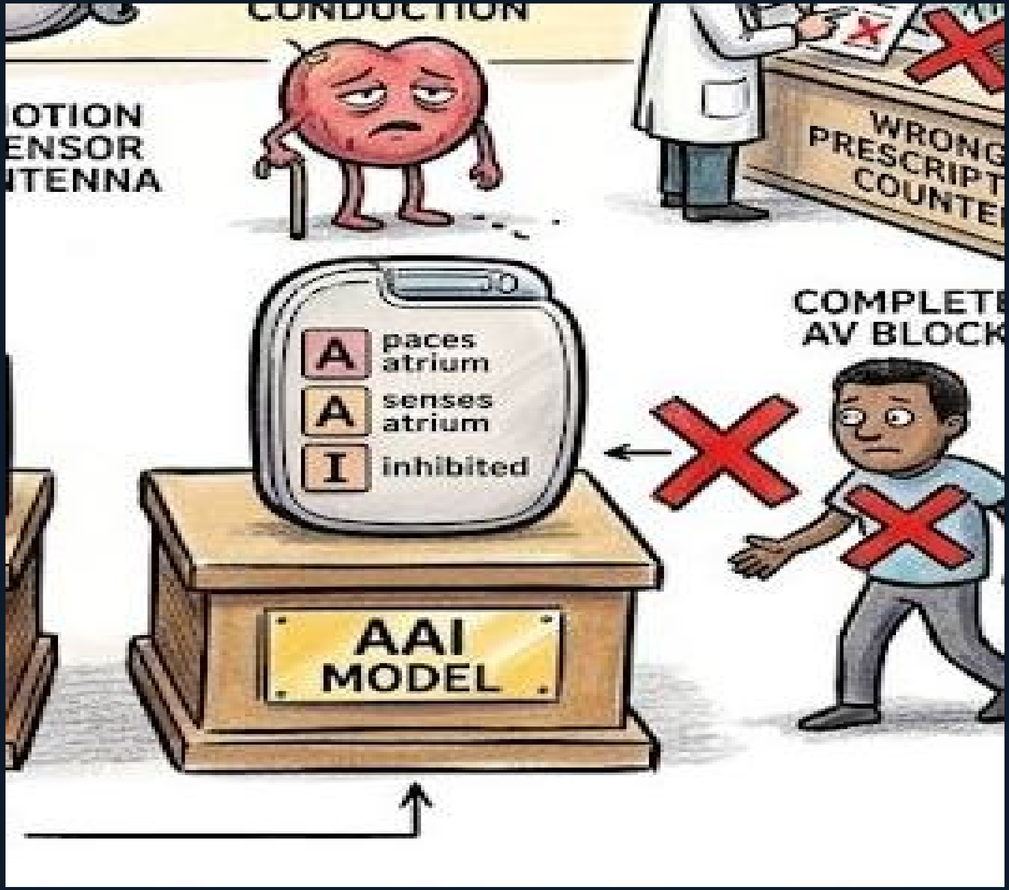


VVIR — Correct for Permanent AF

The VVIR model with a motion sensor antenna receives the permanent AF patient with a thumbs-up and gold star. The code explains itself: V=paces ventricle only, V=senses ventricle only, I=inhibited by sensed beats, R=rate-responsive sensor.

VVIR paces the ventricle at an appropriate rate regardless of atrial activity. The rate-responsive feature (R) uses an activity sensor (accelerometer or minute ventilation) to increase the pacing rate during exercise — compensating for chronotropic incompetence.

VVIR is the correct mode for: permanent AF + AV block, or any patient in permanent AF needing ventricular pacing.



AAI — Sick Sinus + Intact AV Conduction

The AAI model accepts a sick sinus syndrome patient (tired sinus node with cane). Code: A=paces atrium, A=senses atrium, I=inhibited. But when a complete AV block patient approaches, a red X blocks them.

AAI is used for sick sinus syndrome where the AV node and His-Purkinje system are intact. The pacemaker only paces the atrium — the native AV conduction system carries the impulse to the ventricles normally.

Board trap: choosing AAI for a patient with complete AV block — wrong. AAI cannot pace the ventricle, so it is useless when AV conduction is blocked.

Scene H — Board Trap Master Summary

DDD = most physiologic mode

Paces A+V, senses A+V, tracks P waves

DDD + permanent AF = DANGEROUS

Tracks fibrillatory waves → rapid ventricular pacing

VVIR = correct mode for permanent AF

Paces ventricle only; rate-responsive

AAI = sick sinus with intact AV conduction

Atrium only; useless in complete AV block

R = rate-responsive (4th position)

Activity sensor increases pacing rate with exercise

Position 1: chamber paced (A/V/D)

What gets paced

Position 2: chamber sensed (A/V/D/O)

What is detected

Position 3: response (I/T/D/O)

How pacemaker responds to sensed event

Wrong order: DDD for AF patient

Always rejected; use VVI or VVIR

AAI useless in AV block

Cannot pace ventricle when AV node fails